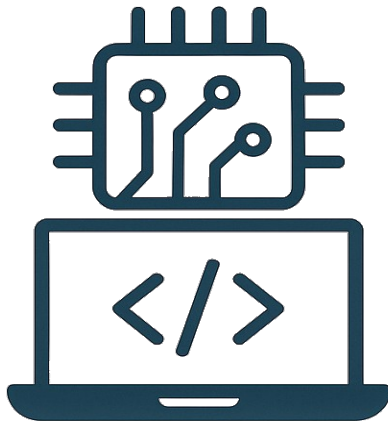




Motorcycle Chain Oiler v. 2

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1. Preface

“Motorcycle Chain Oiler v. 2” was designed as a controller for a motorcycle chain lubrication system. The controller is connected to the vehicle’s electrical system and to a peristaltic oil pump. The selection and construction of the mechanical components are left to the user. Motorcycle riders who do not feel confident building such an oiler themselves should instead use ready-made lubrication systems or alternative lubrication methods.

2. How It Works

A motorcycle chain requires regular maintenance. It should be cleaned and lubricated. In the simplest case, it is cleaned using cleaning spray and a brush, then warmed up by riding a few kilometers, and afterward lubricated with chain spray. All of this not only takes time and money, but also impacts the environment.

A chain oiler can be filled with environmentally friendly oil, is a one-time investment, and operates automatically.

After the ignition voltage is switched on, the microcontroller starts up, reads the settings from non-volatile memory, and waits for the configured time (TD) before the first lubrication cycle. This prevents possible oil dripping while mounting the motorcycle in one’s own yard. After this delay, lubrication is performed — the pump is powered with the configured output level (PW) for the configured duration (TO). Afterward, the pause time (TP) elapses and lubrication is repeated. The parameters must be determined experimentally.

Manual lubrication can also be performed. The pump power level is the same as during automatic operation (PW). The manual lubrication time (TH) is intended for activation while the wheel is rotating. The manual function is also very useful for filling the entire system.

The onboard voltage can also be measured. For this purpose, the voltages of 12.0 V and 7.0 V must first be calibrated.

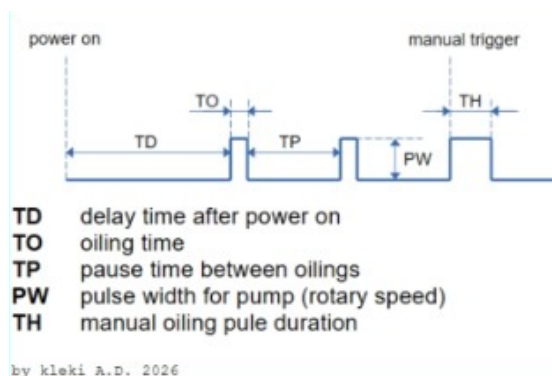


Figure 1: Overview of Parameters

3. Connection to the Vehicle Electrical System

The connection to the vehicle electrical system is made via connector J1. The negative terminal (PCB corner, brown wire) should be connected to the negative terminal of the vehicle electrical system (typically the vehicle ground). The positive terminal (red wire, C2-side, PCB-middle) should be connected, through a fuse, to DIN 72552 terminal 15 (switched positive from the ignition starter switch) of the vehicle electrical system. The maximum permissible current is 2 A. The supply voltage should be between 6 and 15 V.

4. Oil pump

4.1 Oil supply

A peristaltic pump (hose pump) using 4×2 mm silicone tubing and a suitable operating voltage is recommended. An additional brake fluid reservoir, for example, can be used as the oil container. In the reference implementation, such a reservoir was installed under the seat. In example setup, 50 ml of chain oil is sufficient for approximately 1,000 km.

4.2 Electrical connection of the pump

The connection to the oil pump is made via connector J2. The negative terminal (PCB corner, white wire) should be connected to the negative terminal of the pump. The positive terminal (green wire) should be connected to the positive terminal of the pump. Please strictly observe the pump manufacturer's specifications regarding the direction of rotation. The maximum continuous pump current is 250 mA and may be overloaded up to 1000 mA for short periods (up to 1000 ms).

4.3 Chain lubrication

A cannula (syringe needle) size 21G (0.8 mm) is used as the dosing nozzle. Such needles can be purchased at a pharmacy. The tubing consists of 4 mm pneumatic lines combined with sections of rubber hose. **Drilling holes in the rear swingarm is strictly prohibited!**

5. Android-App

An Android app was developed to control the chain oiler. It is used for configuration and manual operation. For everyday use, the app is not required. The oiler operates fully standalone.

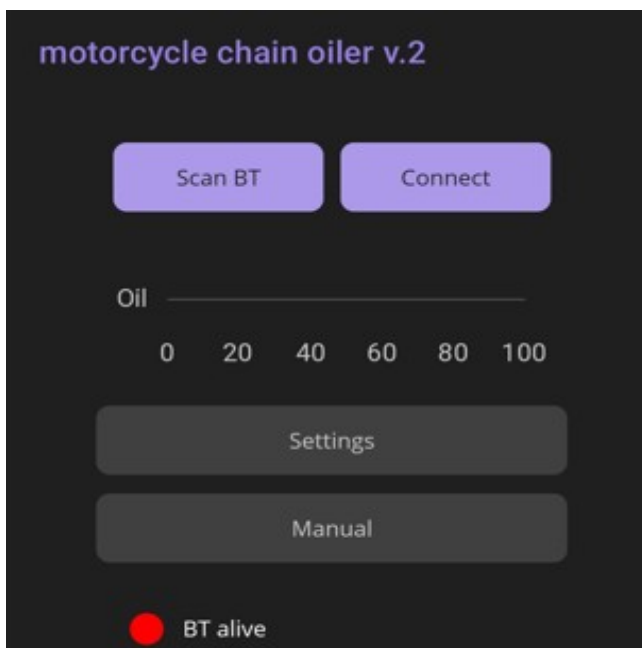


Figure 2: Main window

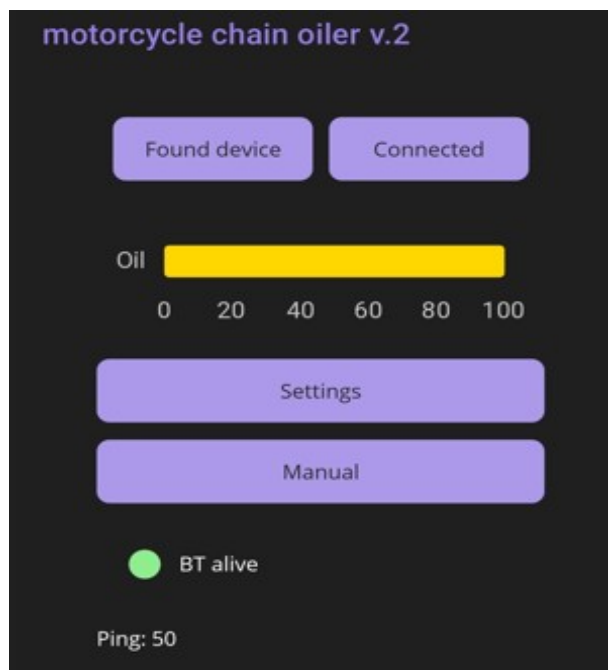


Figure 3: Main window connected

Figure 2 shows the main window. After switching on the ignition voltage, “Scan BT” is pressed, and once a device is found, “Connect” is selected. After a few seconds, the device is ready for operation.

Figure 3 also shows the main window, but in this case the device is already connected to the app.

Yellow bar-graph shows available oil in reservoir.

Pressing “Settings” opens the settings menu, while “Manual” opens the manual mode window (automatic lubrication continues to run).

The “BT alive” indicator informs the user that a connection to the device is active. The “Ping: 132” field provides additional information or continuously changing values. This makes it clearly visible that the oiler is online.

actual values	value to set	set value
10 %	10	set PWM
120 s	120	set TD
55 s	55	set TP
0.2 s	0.2	set TO
10.0 s	10	set TH

BT alive

25 k 25 set VL

RESET OIL

ESP32 got: RE=0 help

Figure 4: Parameters

Figure 4 shows the settings. Their meaning has already been described in section 2. The left column displays the currently set value, while the middle column allows the value to be changed (observe the limits – see section 6.1).

The Container Volume (VL) setting enables the calculation of the remaining oil quantity. The unit used here is Power (PW) multiplied by Pump Time (TO), accumulated over time. When entering the value, it is specified in thousands (for example, 25 corresponds to 25000). The value must be determined experimentally.

The field “ESP32 got: RE=0” shows the communication status between the app and the oiler.

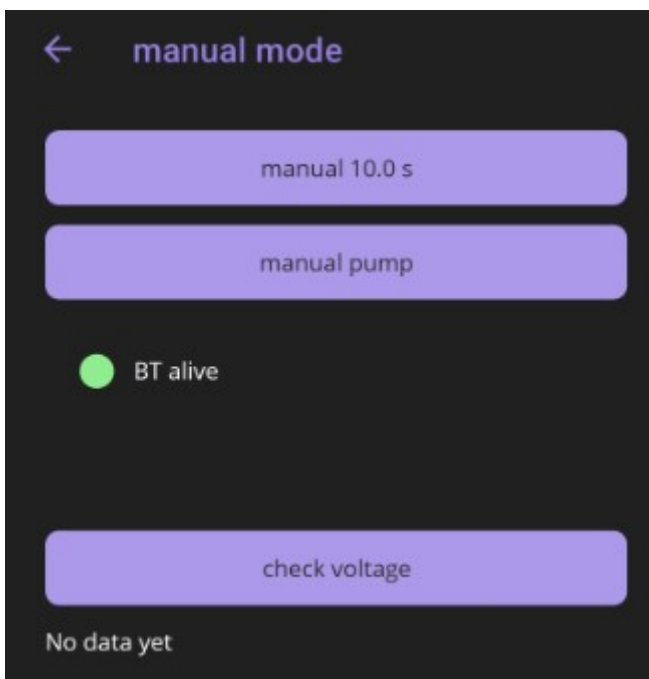


Figure 5: Hand mode

After pressing “check voltage”, two additional buttons appear. These are used for calibrating the voltage measurement and are not protected by any additional prompts or confirmations. Therefore, they must be used with caution.

The measured onboard voltage is displayed in the bottom line of the screen (e.g., “board: -221.1V”). In the example shown in Figure 6, the voltage value is not meaningful — this type of reading is caused by incorrect or missing calibration.

To calibrate the voltage measurement, a laboratory power supply is required. The board is first supplied with 12.0 V and the “12.0V calibration” button is pressed. Afterwards, the voltage is set to 7.0 V and the “7.0V calibration” button is pressed.

Figure 5 shows the manual mode. The “Manual: 10.0 s” button starts manual lubrication for the programmed time “TH”. The process can be interrupted by briefly tapping “manual pump”. The last button is used to manually activate the pump for as long as the button is held. The configured supply power “PW” applies.

The “check voltage” button is used to verify the onboard voltage, provided it has been calibrated beforehand.

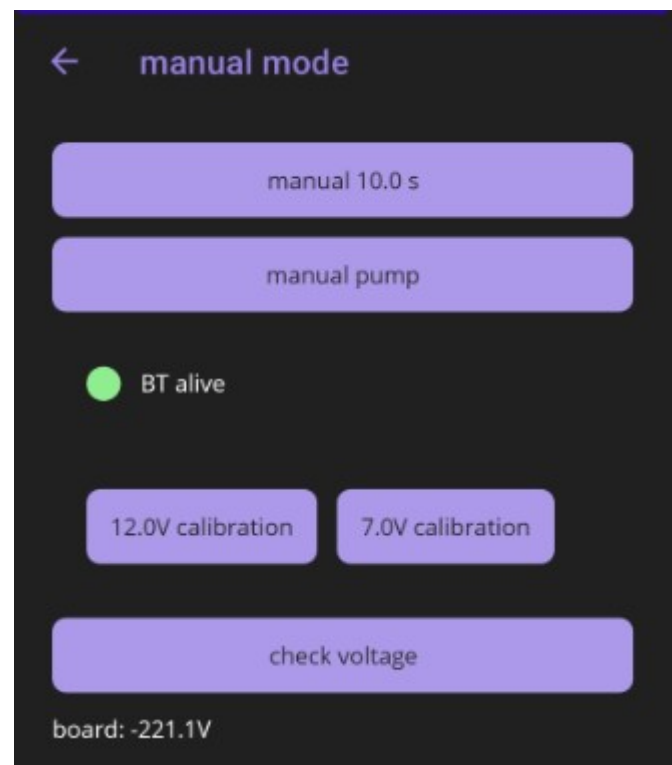


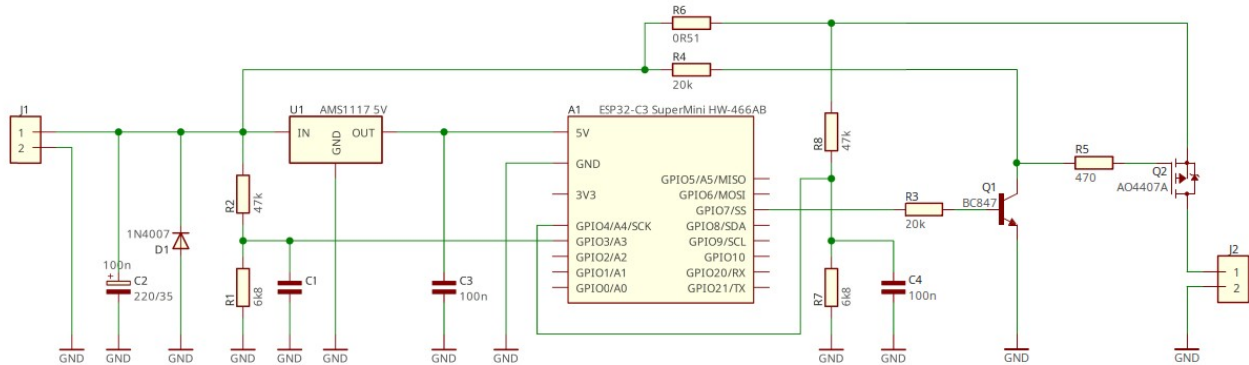
Figure 6: Hand mode with voltage

6. Technical details

6.1 Command list

Command	Example	Limits	Description
PW	PW	-	Query pump control pulse width
PW=	PW=35	0 – 100 [%]	Set pump control pulse width
TD	TD	-	Query lubrication start delay after ignition is switched on
TD=	TD=60	10 – 600 [s]	Set lubrication start delay after ignition is switched on
TP	TP	-	Query pause time between two lubrication cycles
TP=	TP=30	10 - 600	Set pause time between two lubrication cycles
TO	TO	-	Query lubrication duration (pump activation time)
TO=	TO=1,2	0 – 50,0	Set lubrication duration (pump activation time)
TH	TH	-	Query manual activation time for hand lubrication
TH=	TH=2,5	0 – 100,0	Set manual activation time for hand lubrication
MN	MN	-	Start manual lubrication with preprogrammed time
MN=	MN=	0 - 100	Start manual lubrication with specified time
MN=0	MN=0	0	Stop manual lubrication
V12	V12	-	Calibration of supply voltage at 12.0 V
V07	V07	-	Calibration of supply voltage at 7.0 V
VOO	VOO	-	Query voltage value
RES	RES	-	Reset the oil consumption meter
VL=	VL=100000	1000 – 40000000	Set the volumetric capacity of the oil reservoir. The value is time pumped multiplied by pumping power

6.2 Schematics



6.3 Remarks

Even a calibrated voltage measurement is only intended as a rough reference. To determine the battery condition or measure the charging voltage, suitable measuring instruments should be used.

Since the Bluetooth connection is quite susceptible to interference, it is recommended to connect a capacitor with a capacitance of 100 to 330 nF and a breakdown voltage of at least 63 V in parallel with the pump (ideally directly at its terminals).

The blue LED on the PCB is switched together with the pump, allowing quick verification of whether the pump is currently supposed to be running.

6.4 Limits

Supply voltage	6 – 15 V _{DC}
Supply current	2000 mA _{MAX}
Pump voltage	5 – 12 V
Pump current	250 mA _{dauer} , bis 1000 mA _{1000ms}
Environment	-20 – 40°C, 0 – 80 % RH (w/o condensation)

7. GitHub Repository

The complete documentation, including the software, can be downloaded from <https://github.com/klekiMuendling/Motorcycle-chain-oiler-v2>. The Android app is provided as an APK file; during installation, the Android system's security mechanisms must be temporarily disabled. The microcontroller software is provided as an INO file (Arduino IDE).

8. Disclaimer and Notes

This description is provided for orientation purposes only and does not constitute an operating manual. Operational details and correct usage must be determined by the end user. No guarantee is given for the accuracy of the information. Any liability for personal injury or property damage resulting from the operation of the system is expressly excluded. The project and documentation have been created to the best of knowledge and belief; however, errors cannot be completely ruled out.

In the event of a fault, operation must be stopped immediately and the control unit must be professionally taken out of service without delay.

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